

Bicol University Polangui

Polangui, Albay

APPLICATION DEVELOPMENT AND EMERGING TECHNOLOGIES

S.Y. 1st Semester, 2025-2026

FINAL PROJECT AND EXAM:

Automated Web Application Testing Using Selenium

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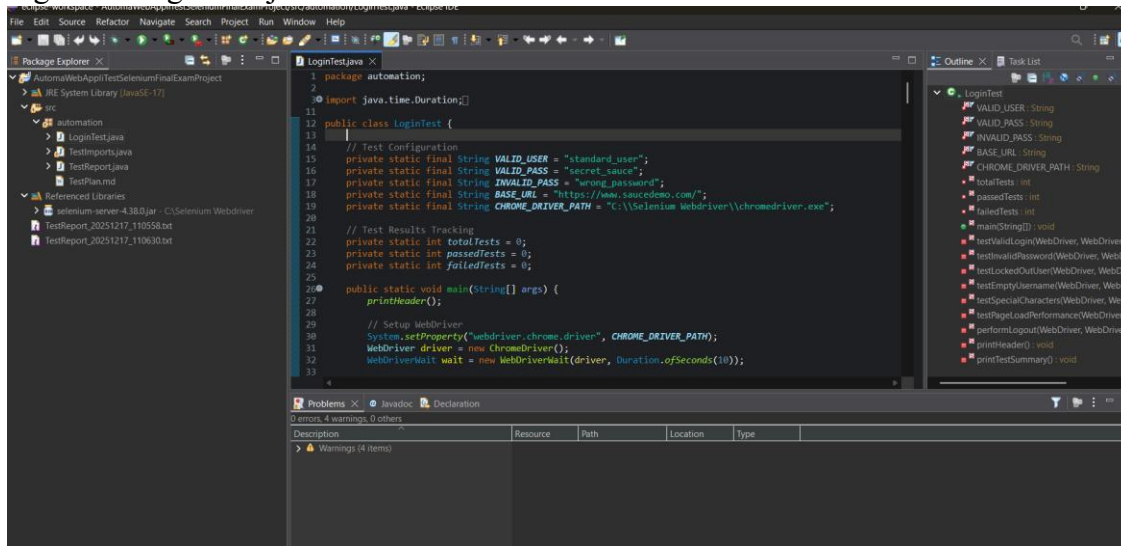
Submitted by:

BSCS2B

Course/Year/Section:

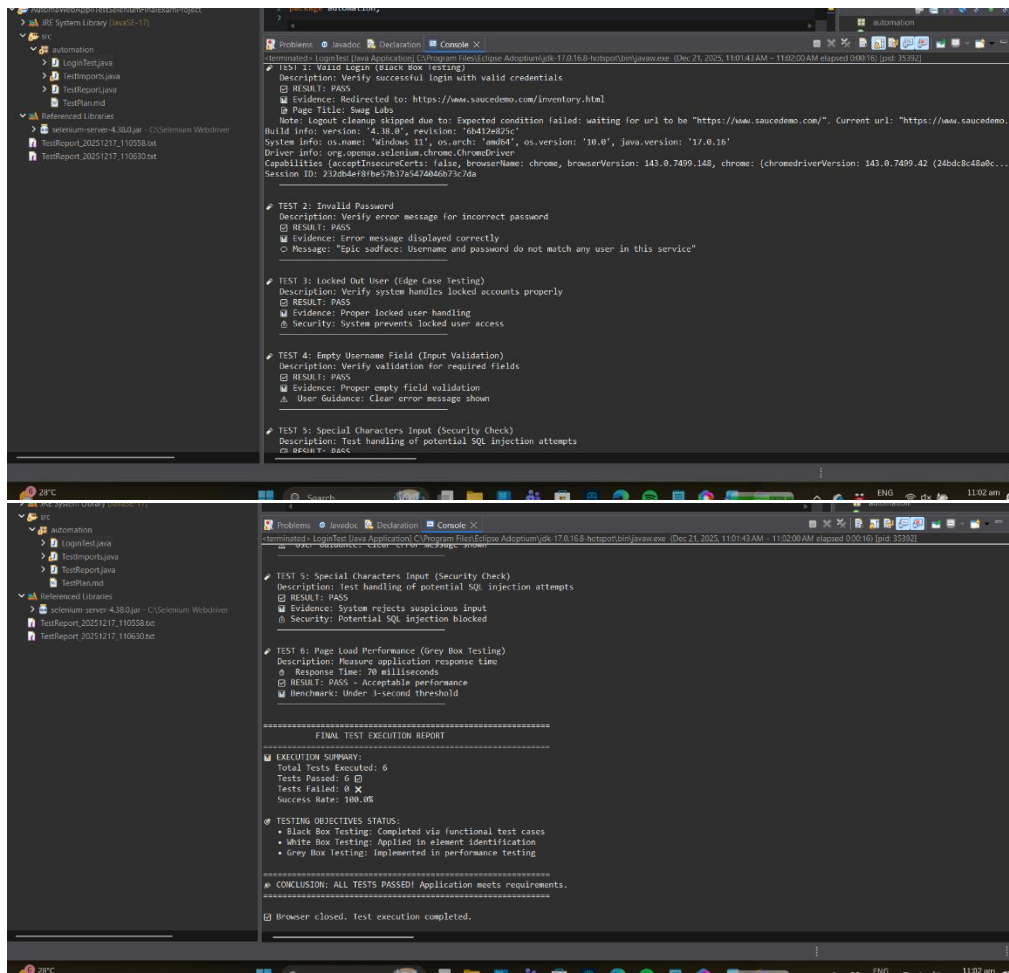
DECEMBER, 19 2025

Figure 1: LoginTest.java code



```
1 package automation;
2
3 import java.time.Duration;
4
5
6
7
8
9
10
11
12 public class LoginTest {
13
14     // Test Configuration
15     private static final String VALID_USER = "standard_user";
16     private static final String VALID_PASS = "secret_sauce";
17     private static final String INVALID_PASS = "wrong_password";
18     private static final String BASE_URL = "https://www.saucedemo.com/";
19     private static final String CHROME_DRIVER_PATH = "C:\\Selenium WebDriver\\chromedriver.exe";
20
21     // Test Results Tracking
22     private static int totalTests = 0;
23     private static int passedTests = 0;
24     private static int failedTests = 0;
25
26     public static void main(String[] args) {
27         printHeader();
28
29         // Setup WebDriver
30         System.setProperty("webdriver.chrome.driver", CHROME_DRIVER_PATH);
31         WebDriver driver = new ChromeDriver();
32         WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
33     }
34 }
```

Figure 2: LoginTest.java output



```
chromedriver: LoginTest Java Application C:\Program Files\Eclipse Adoptium\jdk-17.0.16-hotspot\bin\java.exe (Dec 21, 2025, 11:01:43 AM - 11:02:00 AM elapsed 00:01:16 [pid: 15352])

TEST 1: Valid Login (Black Box testing)
Description: Verify successful login with valid credentials
RESULT: PASS
Evidence: Redirected to: https://www.saucedemo.com/inventory.html
Page Title: Snag Labs
Note: Login attempt skipped due to: Expected condition failed: waiting for url to be "https://www.saucedemo.com/", Current url: "https://www.saucedemo.com/"
Build info: version: "4.18.0", revision: "6b1262f6c"
System info: os.name: "Windows 11", os.arch: "amd64", os.version: "10.0", java.version: "17.0.16"
Driver info: org.openqa.selenium.chrome.ChromeDriver
Capabilities {acceptInsecureCerts: false, browserName: chrome, browserVersion: 143.0.7499.148, chrome: {chromedriverVersion: 143.0.7499.142 (24bdc8c48abc...,
Session ID: 273b0de18be57b57d547d0d0b73c7da

TEST 2: Invalid Password
Description: Verify error message for incorrect password
RESULT: PASS
Evidence: Error message displayed correctly
Message: "Login failed: Username and password do not match any user in this service"

TEST 3: Locked Out User (Edge Case Testing)
Description: Verify system handles locked accounts properly
RESULT: PASS
Evidence: Proper locked user handling
Security: System prevents locked user access

TEST 4: Empty Username Field (Input Validation)
Description: Verify validation for required fields
RESULT: PASS
Evidence: Proper empty field validation
User Guidance: Clear error message shown

TEST 5: Special Characters Input (Security Check)
Description: Test handling of potential SQL injection attempts
RESULT: PASS
Security: Potential SQL injection blocked

TEST 6: Page Load Performance (Grey Box Testing)
Description: Measure application response time
Response Time: 70 milliseconds
RESULT: PASS - Acceptable performance
Benchmark: Under 3-second threshold

=====
FINAL TEST EXECUTION REPORT
=====
EXECUTION SUMMARY:
Total Tests Executed: 6
Tests Passed: 6
Tests Failed: 0
Success Rate: 100.0%

TESTING OBJECTIVES STATUS:
- Black Box Testing: Completed via functional test cases
- White Box Testing: Applied via element identification
- Grey Box Testing: Implemented in performance testing

CONCLUSION: ALL TESTS PASSED! Application meets requirements.

Browser closed. Test execution completed.
```

Figure 3. TestImport.java code

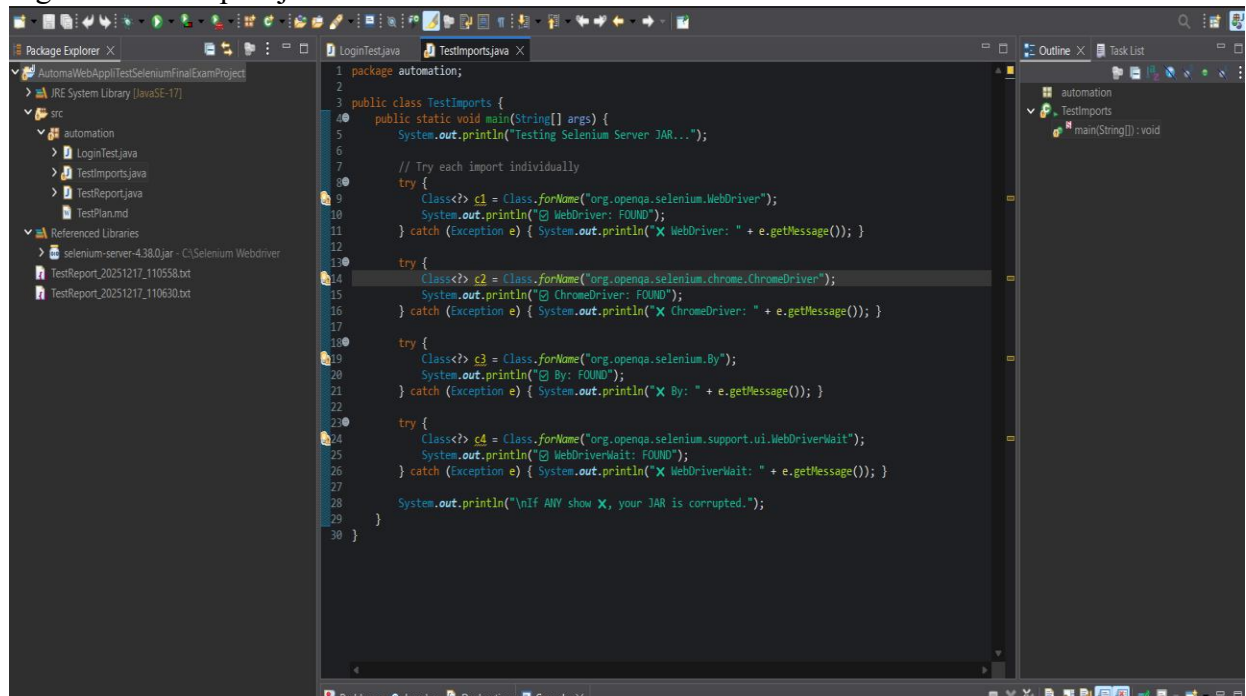


Figure 4. TestImport.java output

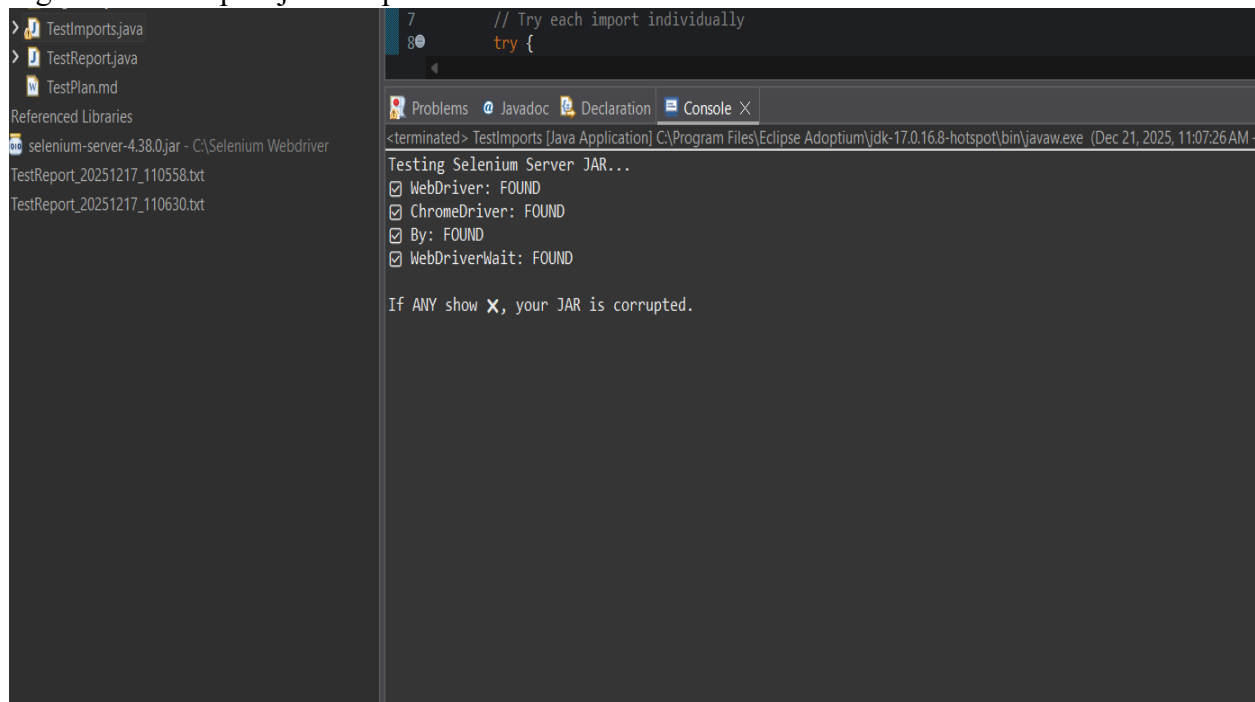
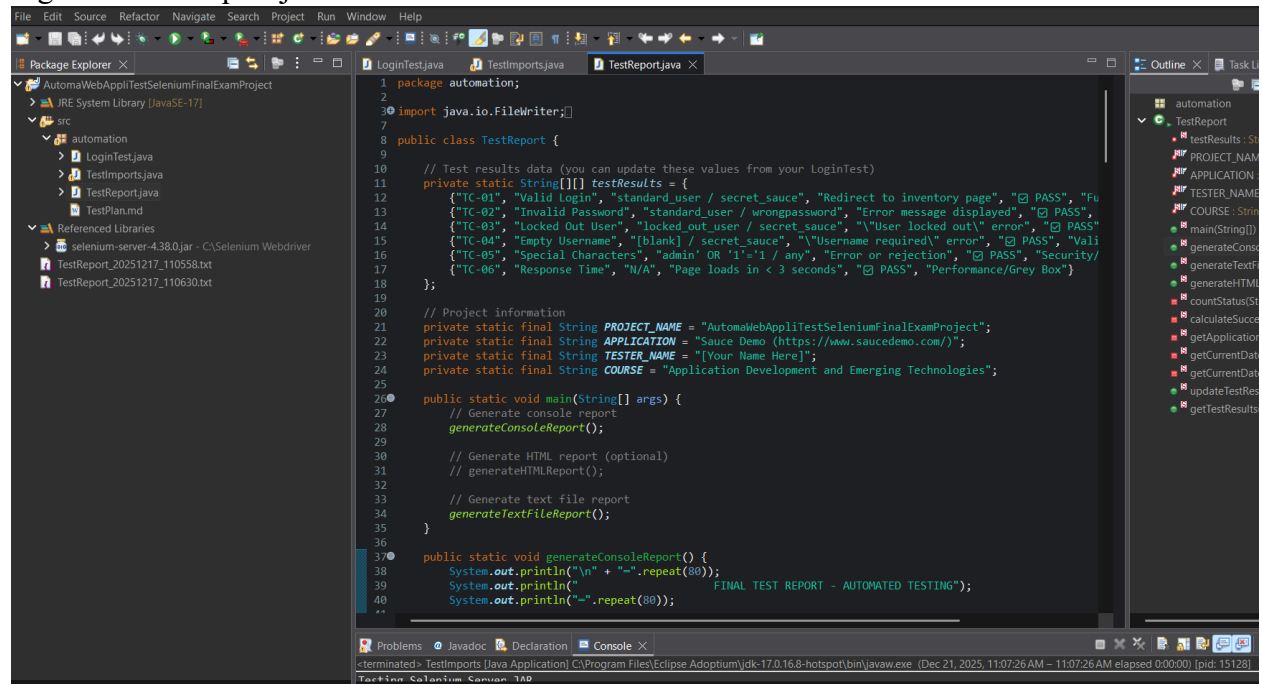
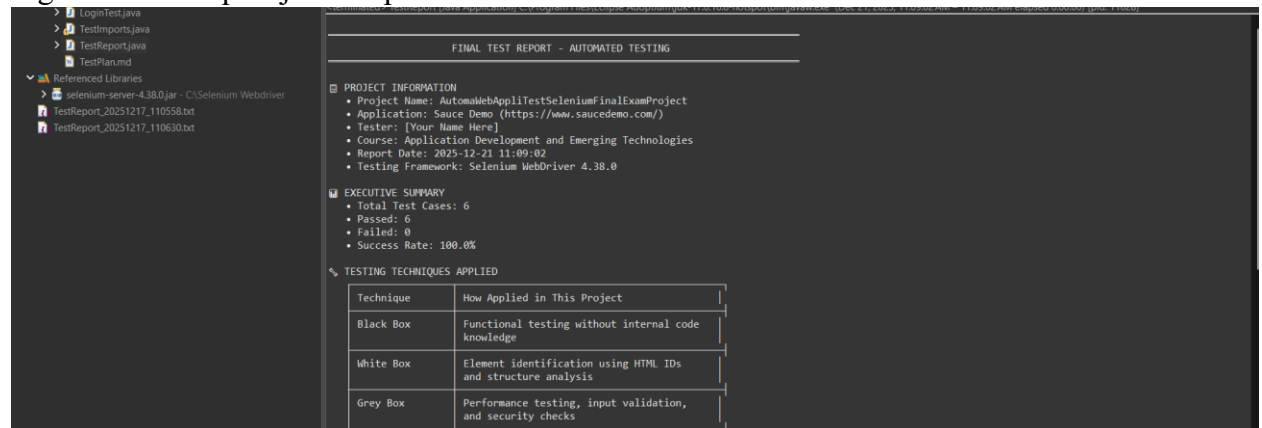


Figure 5. TestReport.java code



```
1 package automation;
2
3 import java.io.FileWriter;
4
5 public class TestReport {
6
7     // Test results data (you can update these values from your LoginTest)
8     private static String[][] testResults = {
9         {"TC-01", "Valid Login", "standard_user / secret_sauce", "Redirect to inventory page", "PASS", "Functional/Black Box"},
10        {"TC-02", "Invalid Password", "standard_user / wrongpassword", "Error message displayed", "PASS", "Functional/Black Box"},
11        {"TC-03", "Locked Out User", "locked_out_user / secret_sauce", "User locked out error", "PASS", "Edge Case/Grey Box"},
12        {"TC-04", "Empty Username", "[blank] / secret_sauce", "Username required error", "PASS", "Validation/White Box"},
13        {"TC-05", "Special Characters", "admin' OR '1'='1 / any", "Error on rejection", "PASS", "Security/Grey Box"},
14        {"TC-06", "Response Time", "N/A", "Page loads in < 3 seconds", "PASS", "Performance/Grey Box"}
15    };
16
17     // Project information
18     private static final String PROJECT_NAME = "AutomateWebApplTestSeleniumFinalExamProject";
19     private static final String APPLICATION = "Sauce Demo (https://www.saucedemo.com/)";
20     private static final String TESTER_NAME = "[Your Name Here]";
21     private static final String COURSE = "Application Development and Emerging Technologies";
22
23     public static void main(String[] args) {
24         // Generate console report
25         generateConsoleReport();
26
27         // Generate HTML report (optional)
28         // generateHTMLReport();
29
30         // Generate text file report
31         generateTextFileReport();
32     }
33
34     public static void generateConsoleReport() {
35         System.out.println("\n" + "-".repeat(80));
36         System.out.println("FINAL TEST REPORT - AUTOMATED TESTING");
37         System.out.println("-".repeat(80));
38     }
39 }
```

Figure 6. TestReport.java output



```
FINAL TEST REPORT - AUTOMATED TESTING

PROJECT INFORMATION
- Project Name: AutomateWebApplTestSeleniumFinalExamProject
- Application: Sauce Demo (https://www.saucedemo.com/)
- Tester: [Your Name Here]
- Course: Application Development and Emerging Technologies
- Report Date: 2025-12-21 11:00:02
- Testing Framework: Selenium WebDriver 4.38.0

EXECUTIVE SUMMARY
- Total Test Cases: 6
- Passed: 6
- Failed: 0
- Success Rate: 100.0%

TESTING TECHNIQUES APPLIED

| Technique | How Applied in This Project |
| Black Box | Functional testing without internal code knowledge |
| White Box | Element identification using HTML IDs and structure analysis |
| Grey Box | Performance testing, input validation, and security checks |

DETAILED TEST RESULTS

| ID | Test Scenario | Input Data | Expected Result | Status | Testing Type |
| TC-01 | Valid Login | standard_user / secret_sauce | Redirect to inventory page | PASS | Functional/Black Box |
| TC-02 | Invalid Password | standard_user / wrongpassword | Error message displayed | PASS | Functional/Black Box |
| TC-03 | Locked Out User | locked_out_user / secret_sauce | User locked out error | PASS | Edge Case/Grey Box |
| TC-04 | Empty Username | [blank] / secret_sauce | Username required error | PASS | Validation/White Box |
| TC-05 | Special Characters | admin' OR '1'='1 / any | Error on rejection | PASS | Security/Grey Box |
| TC-06 | Response Time | N/A | Page loads in < 3 seconds | PASS | Performance/Grey Box |

DEFECT SUMMARY
- No defects found. All test cases passed successfully.

TEST ENVIRONMENT
- Operating System: Windows 11
- Browser: Chrome 143
- Java Version: JDK 17
- IDE: Eclipse
- Automation Tool: Selenium WebDriver 4.38.0
- Driver: ChromeDriver

CONCLUSION & RECOMMENDATIONS
- Application Status: PASS - All test cases passed successfully. Application meets requirements.
- Testing Coverage: Comprehensive - All major login scenarios tested
- Recommendation: Application is ready for deployment
- Next Steps: Expand testing to include other application features

END OF TEST REPORT

Text report saved as: TestReport_20251221_110902.txt
```

Figure 7. TestPlan

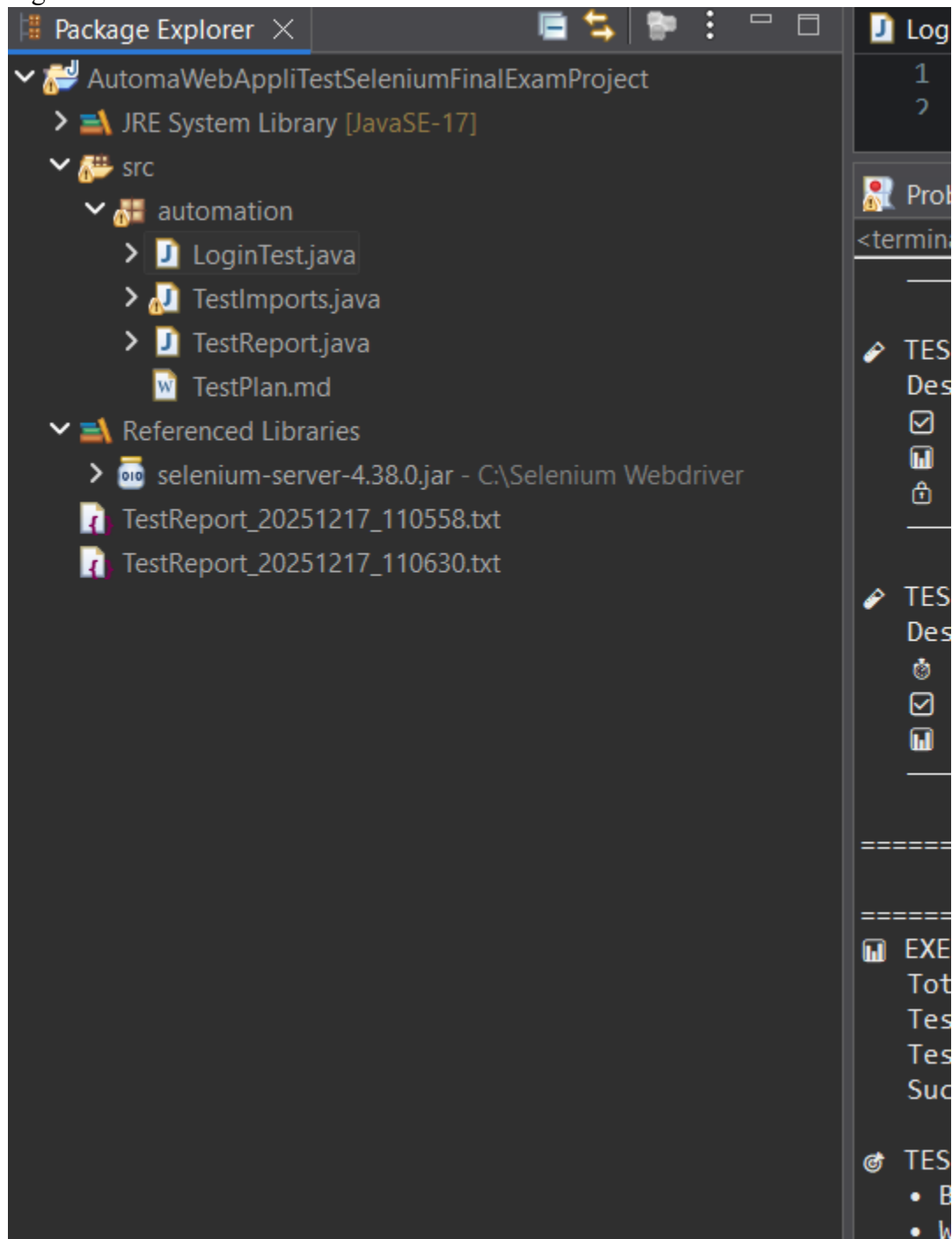
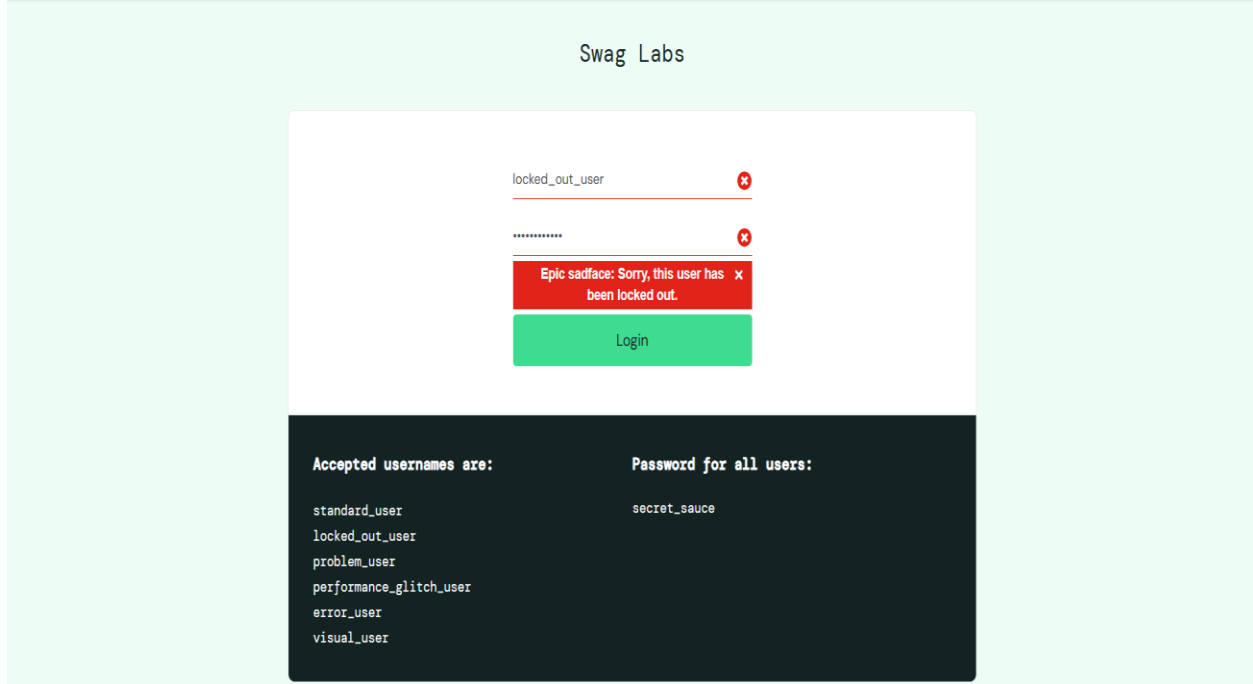
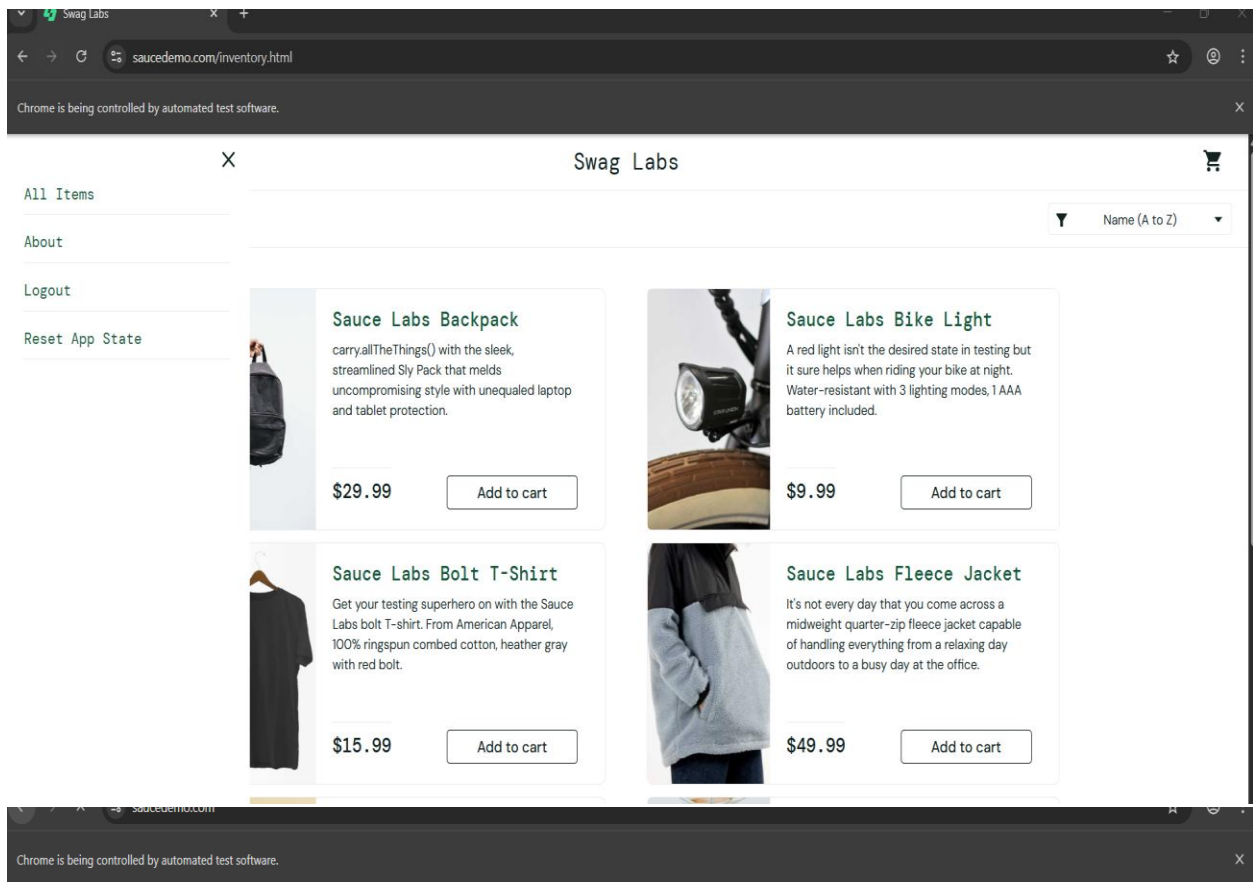


Figure 8. Website automated run test scenario evidence



Reflection:

Working on this Selenium automation project taught me that testing isn't just about finding bugs... It's about thinking like both a developer and an end-user. I started with just a basic understanding of Java, but through trial and error (and plenty of error messages!), I discovered how automated testing tools actually work in real projects.

I learned that Selenium WebDriver isn't a magic wand... it's more like a remote control for browsers. You need to know exactly what buttons to press and in what order. The "black box" testing approach made me think about what users actually experience versus what developers intend. When my test failed because an element wasn't found, I realized: "Oh, this is what users feel when a website doesn't work right!"

The debugging process was humbling. Each error message was a puzzle: "WebDriver cannot be resolved" wasn't just a technical problem, it meant I hadn't properly set up my tools. It reminded me that in software development, 80% of the work is preparation, and 20% is actual coding.

I also discovered that good test cases tell a story. "Valid login" isn't just a checkmark; it's the story of a happy user. "Empty fields" is the story of someone who's distracted or new to technology. "Special characters" is the story of someone testing boundaries. Each test case represents real human behavior.

I tried adding fancy features at first, but they just complicated things. The requirements gave me exactly what I needed: a straightforward approach that anyone could understand. Sometimes in tech, we overcomplicate things when the simple solution is right there. Most importantly, I learned that every error message is an opportunity to understand the system better, and every successful test run feels like solving a mystery.

